

# Omkar Patil

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## Education

### Arizona State University

August 2023 — August 2027

Doctor of Philosophy in Computer Science (3.9/4.0)

### Indian Institute of Technology, Madras

August 2015 — May 2020

Bachelor's in Mechanical Engineering and Master's in Robotics and AI (8.22/10)

## Publications

### Self-Supervised Keyframe Discovery for Horizon-Invariant Behavior Cloning [🔗](#)   December 2026

NeurIPS 2026 / Prabin Kumar Rath, **Omkar Patil**, Nakul Gopalan

Self-supervised method that discovers information-critical observations (mnemonics) as keyframes, conditioning behavior cloning policies on them to retain context over long horizons, without annotations.

### You've Got a Golden Ticket: Improving Generative Robot Policies With A Single Noise Vector [🔗](#)   March 2026

arXiv / **Omkar Patil**, Ondrej Biza, Thomas Weng, Karl Schmeckpeper, Wil Thomason, Xiaohan Zhang, Kausik Sivakumar, Robin Walters, Nakul Gopalan, Sebastian Castro, Stephen Hart, Eric Rosen

Search method to find a single constant noise vector (golden ticket) that improves pretrained diffusion/flow matching robot policies including VLAs (GR00T N1.5,  $\pi 0$ , SmolVLA) without any retraining.

### StageCraft: Execution Aware Mitigation of Distractor and Obstruction Failures in VLA Models [🔗](#)   October 2026

IROS 2026 / Kartikay Pangaonkar\*, Prabin Rath\*, **Omkar Patil\***, Nakul Gopalan

Training-free VLM-based scene staging for VLAs ( $\pi 0.5$ , SmolVLA); +40% absolute success on real robots.

### Factorizing Diffusion Policies for Observation Modality Prioritization [🔗](#)   May 2026

ICRA 2026 / **Omkar Patil**, Prabin Rath, Kartikay Pangaonkar, Eric Rosen, Nakul Gopalan

Method to prioritize observational modalities, such as vision over tactile, for learning diffusion policies.

## Experience

### Research Intern, RAI Institute

July 2025 — January 2026

Post-training of VLAs and generative robot policies on downstream reward functions.

- Developed a latent-steering-based approach that improved generative policies on 46 of 51 tasks without retraining, resulting in the filing of a patent and a publication: <https://lottery-tickets.rai-inst.com/>
- Showed that derivative-free episodic policy improvement can be competitive with policy-gradient and actor-critic methods, especially for VLAs, potentially with a 15 $\times$  reduction in wall-clock time.

### Graduate Research Assistant, Logos Robotics Lab, Arizona State University   August 2023 — Present

PhD research on robot learning for real-world manipulation with Dr. Nakul Gopalan.

- Built ROS 2 teleoperation pipelines (leader arm, VR headset) for a Franka Research 3 arm and a ROBOTIS AI Worker semi-humanoid; collected 500+ demonstrations across 10 tasks.
- Built data-collection and real-time inference pipelines to deploy trained policies, including Diffusion Policy, ACT, and VLAs ( $\pi 0.5$ , SmolVLA), on real robots.
- Fine-tuned policies with behavior cloning and on-robot policy improvement, and deployed simulation-trained policies on hardware (sim-to-real).

### Senior NLP Quantitative Researcher, Wells Fargo

August 2020 — July 2023

- Developed a novel metric to evaluate the quality of LLM-generated paraphrases (arXiv [🔗](#), patent [🔗](#)).
- Surveyed and built LLM-enabled document generation, speeding up internal documentation (arXiv [🔗](#)).

## Service

**Reviewer:** CoRL 2025, 2026; ICRA 2025, 2026; IROS 2025, 2026

**Organizer:** Compositional and Modular Learning in the Era of Scaling in Robotics (IROS 2026) [🔗](#)

## Skills

**Robot Learning:** Imitation Learning, Behavior Cloning, VLA Post-training (Latent Steering, Policy Improvement), Generative Policies (Diffusion, Flow Matching, ACT), Reinforcement Learning (PPO, SAC, Offline RL, On-robot RL, Sim-to-Real)

**Robotics:** Motion Planning (IK, Trajectory Optimization), Perception (Vision, Point Clouds), Control, Hardware (Franka FR3, ROBOTIS AI Worker), Teleoperation (Leader Arm, VR), ROS 2, MuJoCo, Isaac Sim

**Tooling:** VLAs ( $\pi 0.5$ , SmolVLA, GR00T N1.5), LeRobot, PyTorch, JAX, Hugging Face, W&B, Docker, Git